American Economic Review 2024, 114(5): 1382–1414
https://doi.org/10.1257/aer.20220471

Urban Public Works in Spatial Equilibrium: Experimental Evidence from Ethiopia†

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This paper evaluates a large urban public works program randomly rolled out across neighborhoods of Addis Ababa, Ethiopia. We find the program increased public employment and reduced private labor supply among beneficiaries and improved local amenities in treated locations. We then combine a spatial equilibrium model and unique commuting data to estimate the spillover effects of the program on private sector wages across neighborhoods: under full program rollout, wages increased by 18.6 percent. Using our model, we show that welfare gains to the poor are six times larger when we include the indirect effects on private wages and local amenities. (*JEL* H76, I38, J22, J31, O15, O18, R23)

In addition to their direct effects on beneficiaries, social programs can have indirect effects that spill over to nonbeneficiaries and the whole economy. For example, cash and in-kind transfers affect the consumption of nonbeneficiaries and local prices (Angelucci and Giorgi 2009; Cunha, Giorgi, and Jarachandran 2019). Public works, among the most popular forms of antipoverty policy in developing countries, can improve local amenities for beneficiaries and nonbeneficiaries and affect the labor market equilibrium locally and in other locations (Imbert and Papp 2015, 2020).[1] Despite the large literature on social programs, there have been few attempts to fully quantify their effect beyond their direct effects on beneficiaries in targeted

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† Go to https://doi.org/10.1257/aer.20220471 to visit the article page for additional materials and author disclosure statements.

[1] A World Bank (2015) report found public works programs in 94 countries, 39 in sub-Saharan Africa, including large programs in Malawi, Ethiopia, South Africa, and Tanzania.

locations (Egger et al. 2022; Muralidharan, Niehaus, and Sukhtankar 2023), and none in urban areas, where spatial spillover effects are likely to be larger.

Estimating the indirect effects of social programs is challenging for at least five reasons. First, it requires exogenous variation in the implementation of a program on a large scale, which is rare. Second, researchers need information on outcomes of beneficiaries and nonbeneficiaries in treated and untreated locations. Third, when program effects spill over across space, the simple comparison between treated and untreated locations is likely to yield biased estimates and miss benefits to untreated locations.[2] Fourth, the effects of a program once it is fully rolled out may differ from the estimated effects under partial rollout. Finally, a comprehensive program evaluation needs to put together direct and indirect effects in a single metric, e.g., income or welfare.

This paper estimates the direct and indirect effects of Ethiopia's Urban Productive Safety Net Program (UPSNP). The UPSNP is a large urban public works program that offers employment at high wages on small-scale neighborhood projects to poor households for a maximum of three years.[3] Approximately 18 percent of households in the city were eventually enrolled in the program. We exploit the gradual rollout of UPSNP across randomly chosen neighborhoods of Addis Ababa to estimate its short-run effect on earnings and employment, local amenities, and private sector wages among the urban poor. A spatial equilibrium model guides our empirical analysis. We estimate the aggregate wage effects of the program—including spillovers between locations—by characterizing exposure of each location to the program through the commuting network using an expression derived from the model. We also use the model to compute the welfare gains to the poor once the program was fully rolled out, including direct income gains from participation and indirect gains from improvement in amenities and rising wages. Our approach is at the intersection of randomized program evaluation at scale (Muralidharan and Niehaus 2017) and quantitative analysis of spatial equilibrium (Redding and Rossi-Hansberg 2017).

We proceed in six steps. In the first step, we exploit the randomized rollout of the program across neighborhoods (*woredas*) of the city of Addis Ababa in its first year of implementation.[4] We collected precisely geo-referenced panel data on poor households—both beneficiaries and nonbeneficiaries—across the city. We start by comparing households who live in woredas with and without the program.[5] The results suggest that the program generated a large amount of employment on public works, the equivalent of 12.6 percent of hours worked in the control. However, participating households reduced their labor supply to the private sector by about 12.8 percent, so that the net effect of the program on total hours worked is close to 0 and

[2] A typical solution to this problem in rural settings compares among untreated villages those that are within or beyond a certain radius from treated villages (Egger et al. 2022; Muralidharan, Niehaus, and Sukhtankar 2023). This approach is ill suited to urban settings where economic interactions between neighborhoods are strong and not only based on geographic proximity.

[3] The temporary nature of the UPSNP is typical of workfare programs in sub-Saharan Africa (Beegle, Galasso, and Goldberg 2017; Alik-Lagrange et al. 2020; Bertrand et al. 2021).

[4] Treatment woredas were chosen randomly through a public lottery.

[5] We prespecified the experimental design, the labor supply and amenities outcomes, and the treatment versus control specifications in a pre-analysis plan at AEARCTR-0003387 (https://www.socialscienceregistry.org/trials/3387).

insignificant.[6] This reduction in labor supply is likely to affect private sector wages, but since 55 percent of workers commute to another woreda, the wage effects of the program are likely to spill over beyond treated woredas, so comparing wages in treated and control areas would yield unreliable estimates.

To guide our evaluation of the direct and spillover effects of the UPSNP, in the second step we develop a spatial equilibrium model that borrows from the urban economics literature (Heblich, Redding, and Sturm 2020; Balboni et al. 2021).[7] We leverage the model (i) to estimate labor market spillovers across the city; (ii) to quantify the welfare effects of the program including direct benefits, effect on amenities, and labor market effects; and (iii) to provide counterfactual analysis of the program under full rollout and compare it with a cash transfer.

In the third step, we estimate labor market spillovers. The model provides an expression for the equilibrium wage effect in each local labor market as a function of exposure to changes in labor supply from commuters who live in treated neighborhoods. We use rich commuting data to measure wages in each labor market, i.e., in the woreda where workers earn rather than where they live.[8] We then regress wages on exposure to the program for each labor market, which we construct as the sum of treatment status in each place of residence weighted by the share of workers who commute from there. Exposure has a shift-share interpretation, where the shift is the randomly assigned program implementation and the shares are commuting shares at baseline.[9] To account for the fact that even if treatment is randomized, exposure to the treatment is not randomly assigned, we follow Borusyak and Hull (2023) and re-center our measure of exposure using potential exposure to 2,000 re-randomizations of the treatment assignment. Our estimates imply that under partial rollout, private sector wages increased by 14 percent in treated and 3 percent in untreated labor markets and that under full rollout, they increased by 18.6 percent everywhere. By contrast, a simple comparison of wages earned by residents of treated and control woredas that ignores spatial spillovers would imply a much smaller 9.3 percent wage increase.[10]

In the fourth step, we estimate the effect of the program on local amenities. We use an index that aggregates five subjective indicators of amenities that might plausibly have been affected by the program. We estimate an improvement in neighborhood quality by roughly 0.6 standard deviations in treated neighborhoods, perceived

[6] The program increases labor supply at the extensive margin among treated households because some participants would not otherwise have been working. But program participants who would otherwise have been working in the private sector reduce their hours by approximately twice the hours worked on the program, leading to zero net change in hours worked. Households in the program still experience sizable increases in income relative to control households because the program pays wages well above the level in the private sector.

[7] Our model is a simplified version of theirs, as it includes no housing markets or trade. We estimate small and insignificant effects of the program on residential mobility, rents, consumption expenditures, and local prices.

[8] Like Monte, Redding, and Rossi-Hansberg (2018), we do not take a stand on the spatial extent of labor markets. Instead, we define local labor markets as the most fine-grained possible geography (i.e., the woreda) and explicitly model linkages between labor markets. Throughout the paper, "labor market" refers simply to the woreda in which people work.

[9] Our use of a model-based exposure in a reduced-form estimation is reminiscent of the "market access" approach by Donaldson and Hornbeck (2016). Alternatively, we instrument labor supply to each labor market by exposure—which includes endogenous endline commuting probabilities—and obtain consistent estimates.

[10] In online Appendix E we show that alternative approaches to estimating spillovers based on geographical proximity between woredas do not provide reliable estimates. We also check that the wage spillovers do not bias the ITT estimates on employment in online Appendix F.

by both beneficiary and nonbeneficiary households.[11] To quantify the value of improvements in public goods, we correlate these measures of local amenities with private market rents. Overall, we estimate an effect on amenities equivalent to 2.5 percent of total local amenity value.

In the fifth step we use a gravity equation to estimate the Frechet parameter, the key parameter of the model that governs the distribution of the idiosyncratic taste for working in a given location and therefore determines how much urban residents' welfare improves when wages rise in the labor markets they commute to. We estimate the parameter as the elasticity of commuting with respect to wages at destination, which we instrument by the destination's exposure to the program. Our estimate of 2.08 is comparable to estimates by Tsivanidis (2019) for Bogotà and Kreindler and Miyauchi (2023) for Dhaka and Colombo.[12]

Finally, we use the structure of the model to compute the welfare gains to the poor from the program, combining the direct income effects on participating households, equilibrium wage effects, and improvements in local amenities in treated woredas.[13] Our model allows us to consider two scenarios: when the program was partially rolled out and after it was implemented in all woredas. We show that under partial rollout, residents of treated woredas were the ones who gained the most from the program, but residents of control woredas experienced substantial benefits through rising private wages. Under complete rollout, the welfare gains extended to all woredas and became larger, due to equilibrium effects. The welfare of the urban poor increased by 22.4 percent, including a 3.7 percent direct gain from participation to public works and a 16.2 percent gain from rising private sector wages.[14] As a benchmark, we compare these welfare gains to the gains from a cash transfer that pays public works' wages without affecting labor supply. The cash transfer does better when one considers only the direct benefits from participation, but public works dominate once effects on amenities and wages are taken into account.[15]

Our paper is the first to combine a randomized control trial of a social program at scale and a spatial equilibrium model to identify and quantify its direct and indirect effects in the presence of spatial spillovers. As such, it contributes to three main strands of the literature. First, we contribute to the literature on the equilibrium and spillover effects of antipoverty programs using large cluster-randomized controlled trials (Egger et al. 2022; Muralidharan, Niehaus, and Sukhtankar 2023). These papers either assume noninterference between potential treatments units or define exposure to spillovers as a parametric—usually stepwise—function of Euclidean

[11] Because UPSNP projects were carried out on a small scale within treated neighborhoods, we do not expect spillover effects on amenities in control neighborhoods.

[12] Alternatively, in online Appendix G we estimate the Frechet parameter as the elasticity of commuting with respect to commuting costs instrumented by walking distance, as in Heblich, Redding, and Sturm (2020), and find a higher estimate, similar to theirs. We provide welfare calculations based on this alternative estimate and check that the conclusions are unchanged.

[13] We focus our welfare calculations on poor households who are the target of the program. Richer households do not participate in the program but pay taxes to fund it and may benefit from improved amenities or suffer from having to pay higher wages as employers (Imbert and Papp 2015; Muralidharan, Niehaus, and Sukhtankar 2023).

[14] The welfare analysis does not include changes in prices or rents; we find no short-term impact of the program on household consumption or local prices, and most of the urban poor live in government housing and do not pay rent.

[15] In online Appendix H, we show that public works still do better than cash in terms of income gains, i.e., if we do not use the structure of the model, ignore amenities, and focus on gains from direct participation in public works and rising private wages.

distance to treated areas. While this assumption may be justified in the context of relative remote rural villages, it is unlikely to hold in urban areas that are closely connected by commuting between labor markets. In fact, the "donut" or "radius" approaches used in other papers provide insignificant and unstable estimates in our setting. Our model-based approach allows us to estimate spatial spillovers in a network of locations linked by commuting flows under partial and full rollout. In doing so, our paper provides a new answer to the long-standing question of how to use randomized control trials to quantify the effect of policies at scale (Deaton 2010; Muralidharan and Niehaus 2017; Bergquist et al. 2019).

Second, we provide a new application of spatial equilibrium models to the empirical analysis of urban change. Most papers study variations in commuting costs due to changes in the transportation network in historical cities (Heblich, Redding, and Sturm 2020; Ahlfeldt et al. 2015) and cities in developing countries today (Tsivanidis 2019; Balboni et al. 2021). Instead, in our application to urban public works programs, we estimate the effects of changes in labor supply through the existing transport network. We borrow from other papers (Heblich, Redding, and Sturm 2020; Balboni et al. 2021) to model commuting decisions, the spatial labor market equilibrium, and the welfare effects of changes in wages and amenities. Like Balboni et al. (2021), we overcome the challenge of data scarcity that has so far limited the application of these models to cities in developing countries (Bryan, Glaeser, and Tsivanidis 2020) by measuring amenities and wages, as well as commuting flows, costs, and times at the individual level in original survey data. We also improve on identification by exploiting random variation in the placement of the program across neighborhoods. This enables us to estimate the Frechet parameter as the elasticity of commuting with respect to exogenous changes in destination wages driven by exposure to the program. Our estimate is similar to nonexperimental estimates by Tsivanidis (2019) and Kreindler and Miyauchi (2023).

Third, by quantifying equilibrium changes in wages across all locations in the urban network, we relate to the literature on local labor markets, local development policies, and the spatial transmission of labor market shocks (Moretti 2011; Kline and Moretti 2014; Manning and Petrongolo 2017; Monte, Redding, and Rossi-Hansberg 2018; Monras 2020; Imbert and Papp 2020). In particular, Monte, Redding, and Rossi-Hansberg (2018) study equilibrium responses to local labor demand shocks in US commuting zones and emphasize that openness to commuting dissipates the effects of these shocks on local employment. Using a different approach, Manning and Petrongolo (2017) structurally estimate a job search model and find that while the search radius of a given job seeker is small, labor markets largely overlap, so that local shocks are likely to have ripple effects. We contribute to this literature by directly estimating the equilibrium effects of a labor market shock using the randomized program rollout for identification and detailed information on commuting networks. In doing so, we provide some of the first evidence on the spatial extent of labor markets within developing-country cities. Cities in Africa, in particular, have been characterized as having highly fragmented labor markets, based largely on the observation that most workers walk to work (Lall, Henderson, and Venables 2017). Evidence suggests that spatial frictions may be important in these contexts (Franklin 2018; Abebe et al. 2021). We find that despite these frictions, there are substantial commuting flows between neighborhoods, so that a place-based policy that is earmarked for local residents has strong local labor market effects as well as large spillover effects to untreated neighborhoods.[16]

Our paper is also the first to evaluate the welfare effects of a public works program on the urban poor by estimating experimentally improvements in local amenities, equilibrium wage effects, and direct benefits to participants. A large literature has estimated the effects of public works programs on program beneficiaries (Berhane et al. 2014; Bertrand et al. 2021; Beegle, Galasso, and Goldberg 2017; Alik-Lagrange et al. 2017).[17] The study of indirect effects via labor markets and public good provision has proved more challenging. In particular, there is very little evidence on the effect of public works programs on local amenities.[18] Closely related to this paper, Imbert and Papp (2015) and Muralidharan, Niehaus, and Sukhtankar (2023) estimate positive equilibrium effects of India's rural public works program on rural wages, and Imbert and Papp (2020) estimate spillovers on urban areas due to changes in seasonal migration flows. As compared to these papers, ours combines the advantage of random program placement, measures of local amenities, a mapping of spatial interactions, and a spatial equilibrium model to estimate labor market spillovers under partial and complete rollout. We make progress toward a comprehensive evaluation of public works programs by including direct and indirect effects in a model-based welfare analysis.[19]

The paper proceeds as follows. Section I describes the program, the evaluation design, and the economic lives of program beneficiaries. Section II estimates the effect of the program on employment and amenities, which motivate the model we develop in Section III. Section IV uses the model to quantify the effect of the program on labor markets and welfare. Section V concludes.

I. Program and Data

A. *Program*

The Urban PSNP takes its name from the PSNP (Productive Safety Nets Program), which has been running throughout rural Ethiopia since 2005 (Berhane et al. 2014). The UPSNP was introduced in 2017 in 11 cities in the country (1 city from each region and chartered city) and provides guaranteed public work to targeted households. We evaluate the program in the capital city of Addis Ababa, a city of an estimated 5.2 million people. At full scale, 18 percent of households in the city were enrolled in the program, which translates into 70 percent of all UPSNP beneficiaries in the country. In what follows, we describe the rollout and beneficiaries for Addis

[16] An additional contribution of our paper is to test and reject the "surplus labor" hypothesis in an urban setting. In a world of "surplus labor," hiring workers should have no effect on private sector employment or wages (Lewis 1954; Harris and Todaro 1970). Breza, Kaur, and Shamdasani (2021) provide experimental evidence of this in the lean season in rural India. By contrast, in our setting public works increase private wages and crowd out private sector one-for-one.

[17] For a comprehensive review of the literature on the effects of India's employment guarantee on economic and social outcomes, see Sukhtankar (2016).

[18] Gazeaud and Stephane (2023) find no effect of the rural PSNP on vegetation cover in Ethiopia.

[19] Our paper considers only the contemporaneous effects of the program. Alik-Lagrange et al. (2017) and Bertrand et al. (2021) evaluate the effects of public employment on labor market outcomes of beneficiaries *after* they leave the program.

Ababa only. The program is implemented by local government administrative units, or *woredas*, within cities.

Public Work and Wages: Each beneficiary household is offered up to 60 days of public works per year per working-age member, up to a maximum of 4 members (a maximum of 240 days of work a year). Households are enrolled into the program for three years in total.[20] This relatively short-lived availability of the program is a common feature of workfare programs. For example, Bertrand et al. (2021) study a program that lasts for only six months. They point out that these programs are often used in response to transient negative shocks. Households are free to choose who within the household will do the work, although those individuals need to have been registered as eligible at the time of the household targeting. Work activities take place for between four and five hours per day, compared to an average of nine hours in the private sector. Conditional on completing the work, households are paid Br60 (around US$2) per day, or Br12 per hour. This compares to Br46.5 per day and Br8.4 per hour in the private sector among eligible households in our baseline data.[21] The average beneficiary household earns roughly Br1,000 (US$33) per month, or 30 percent of average household consumption for eligible households in our data.

All work is done in local communities within the woreda. As a result, most public work takes place very close to beneficiary households' place of living. Program wages are paid at the household level, into bank accounts set up in the name of the head of the household. The work consists of small-scale activities aimed at neighborhood improvement. The most common activities are cleaning streets, maintaining drains and ditches, garbage disposal, and greening of public spaces (planting of trees and gardening). In a few rare cases, the works included construction of small cobbled streets in slum areas. Most program beneficiaries report participating in multiple or all of these activities.

Cash Arm: In addition to the public works, the program includes an unconditional cash transfer arm, known as the "direct support" (DS), aimed at poor households unable to participate in public works due to chronic illness, age, or disability. This transfer is considerably smaller than the wages from public works.[22] Although our study is designed to separately identify the effects of the DS, we do not focus on those results in this paper. We estimate negligible impacts of the DS across a range of outcomes, which makes us confident that this component is not driving the equilibrium effects of the program.

Targeting: Households are selected for the program by local community committees within woredas. A strict residential requirement was enforced: only households

[20] The number of days available to each household decreases incrementally with each year in the program, but this does not occur within the time frame of this evaluation.

[21] We estimate an average hourly wage premium from public works of 1.67. These wages are even more attractive for the lower-earning members of targeted households, who are more likely to take up the public works. Online Appendix Figure A2 shows the distribution of wages paid by public works as compared to private sector wages in the control group.

[22] The DS provides Br170 per person per month; the average household enrolled in DS receives Br350 per month, a third of average monthly public works wages.

that were resident in the woreda for at least six months could be selected for the program. Committees aimed to select households in their communities based on their level of poverty. Wieser, Franklin, and Hari (2021) shows that targeted households are poorer (in terms of asset ownership and housing quality) than representative households below the poverty line in representative household survey data from the same year as the baseline for our evaluation.

Take-Up: Take-up of the program is almost universal: fewer than 3 percent of the households in our sample report being offered the program and declining. Within households, public works is mostly done by women and, in particular, older women. Sixty-eight percent of program participants are over the age of 35 (see online Appendix Figure A1). We also find that individual participation is higher for those with no formal education or only primary school. Online Appendix Table A5 compares the labor market engagement of participants compared to the full sample at baseline: before the program, participants were as likely to be employed as the average adult in the sample. This may seem surprising for a program that hires mostly older women. However, program participants are often heads of poor households who work to support their family. In most households, 1 person did all 240 public works days (the average household has 3.5 working-age adults). At full scale, the UPSNP employed nearly 4 percent of all adults in the city.

B. *Evaluation and Data*

The program was randomized at the woreda level in Addis Ababa. In year 1 of the program, only households residing in woredas with poverty rates above 20 percent were eligible for the program: specifically, 90 out of 116 woredas in the city. Randomization was conducted by a public draw of woreda names in November 2016 and stratified by subcity (ten urban sectors within Addis Ababa). Of these 90 eligible woredas, 35 were randomly selected for the program in year 1 (henceforth, treated woredas) and the other 55 woredas to receive the program in year 2 (control woredas).[23] Figure 1 shows a map of the randomization outcomes at the woreda level. Because of lower poverty rates in the 26 ineligible woredas, 86 percent of the poor population lived in eligible woredas. In year 2, *all* woredas were treated, including the 26 woredas that were not eligible in year 1. These 26 woredas are not included in the evaluation because they were not eligible for randomization in year 1.

We conducted a baseline survey immediately after the randomization of woredas into the program but before targeting and rollout of the program occurred.[24] Our aim was to select a representative sample of poor households, among whom we expected to see eligible households well represented in the targeting. First, we conducted a short screening survey of nearly 30,000 households drawn from a random sample of all households in the city. We used random walk sampling starting from randomly selected points within each of the 90 eligible woredas. We used these data to derive

[23] For the Addis Ababa City Administration, public randomization offered a solution to the questions of fairness and reduced speculation about corruption. The World Bank also saw the advantage of this transparent approach and favored randomization for the purposes of a robust impact evaluation.

[24] We describe the timeline of the rollout in more detail in online Appendix Table A1.

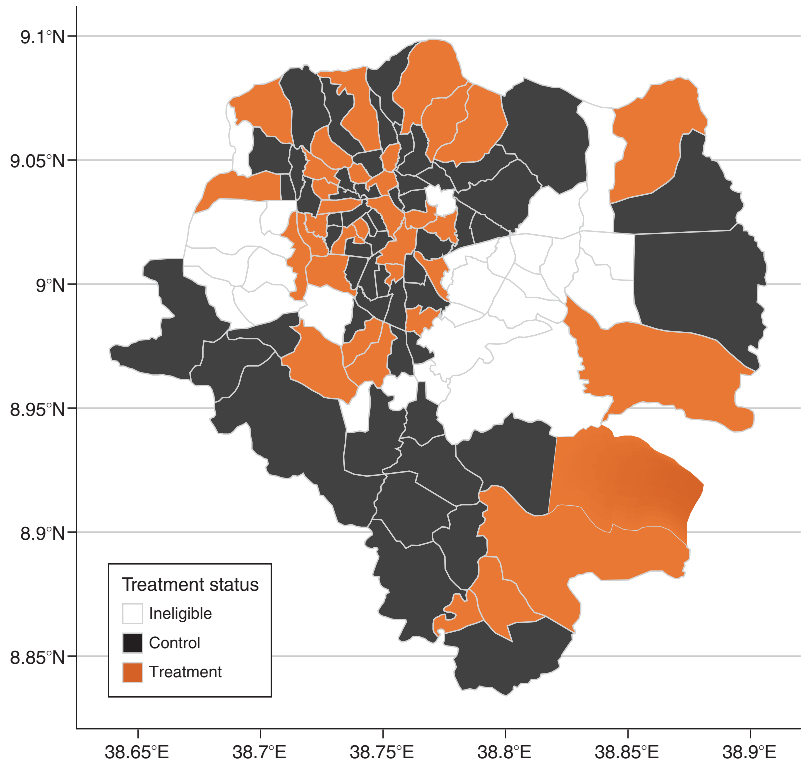

Figure 1. Randomization of the Program across Eligible Woredas

a predicted poverty score using a proxy means test (PMT) for consumption poverty using housing composition, assets, housing, and education. Next, we selected the poorest 28 percent of households in the distribution of PMT scores and conducted a detailed baseline survey. This constitutes our evaluation sample of 6,096 households.

We conducted a detailed endline survey one year later. We found 5,912 out of 6,096 households, an attrition rate of 3.01 percent.[25] Our labor market analysis focuses on all working-age (aged 16 to 65) household members, of which 19,422 were interviewed at endline. We identify within our sample eligible and ineligible households. Throughout the paper, we use *eligibility* to refer to whether a household was selected by the local community regardless of the year in which their woreda was treated.[26] This allows us to estimate the effect of the program on both eligible and ineligible households and individuals using year 1 endline data.[27] Roughly 45 percent of households in our sample are beneficiaries of the public works program, across treatment and control woredas alike. We also followed up with our sample

[25] Online Appendix Table A2 shows that there is no significant difference in attrition rates by treatment across treated and untreated in woredas. Very little else is correlated with response rates; larger households and households in kebele housing are slightly more likely to be found.

[26] For year 2 (control) woredas, we measured this with an additional endline survey conducted a few months after the main endline, when the program had been rolled out in those woredas.

[27] We fail to reject a joint significance test of woreda fixed effects on beneficiary observables, which suggests that the targeting was done in a similar way across woredas in the city.

Table 1—Description of Evaluation Sample at Baseline Compared to Contemporaneous LFS (2018) in Addis Ababa (Working-Age Individuals Only)

	Evaluation sample		LFS 2018	
	Mean	SD	Mean	SD
	(1)		(2)	
Age	32.31	(13.00)	31.71	(11.71)
Share female	0.524	(0.499)	0.555	(0.497)
Share in any employment	0.479	(0.500)	0.561	(0.496)
Available	0.211	(0.408)		
Inactive	0.150	(0.357)		
Studying	0.163	(0.369)	0.142	(0.349)
Unemployed (active search)	0.047	(0.211)	0.060	(0.238)
Share in self-employment	0.110	(0.312)	0.191	(0.393)
Share in wage work	0.380	(0.485)	0.393	(0.488)
Share with permanent wage contract	0.082	(0.275)	0.171	(0.376)
Average working hours a week	21.19	(25.54)	28.20	(28.37)
Average share of 48 weekly hours worked	0.441	(0.532)	0.587	(0.591)
Average monthly wage (wage work only)	1,544.9	(1,272.2)	3,353.8	(3,286.5)
Average hourly wage (wage work only)	9.694	(12.76)	18.15	(20.82)
Commutes out of own Woreda (share of workers)	0.546	(0.498)		
Commuting cost (share of monthly earnings)	0.057	(0.225)		
Commuting time (minutes per day)	48.54	(38.80)		
Observations	20,119		8,577	

Notes: The sample comprises all working-age adults at baseline. Working-age adults are those aged 16 to 65. The labor force survey (LFS) data we use are the Urban Employment and Unemployment Survey (UEUS) conducted by the Ethiopia Statistical Agency in 2018 (for details, see http://www.csa.gov.et/). We define "Inactive" as all workers who are not working and also not available to work. This is largely those who do unpaid work in the home (mostly women) as well as the disabled, retired, or unwilling to work for other reasons. Those "available" to work are those who did not work in the last seven days but said they would work if offered. Roughly a quarter of this group said that they have irregular work or have some attachment to a job that they did not do in the last seven days. We cannot compute the share of nonworkers who are either "available" or "inactive" in the LFS because the survey does not ask about availability. We can compute the share who are unemployed from the module on job search.

another year later. Because all control woredas were in the program by this point, we do not use these data for the labor market analysis.

Tables A3 and A4 in the online Appendix show no sign of imbalance in baseline characteristics between treated (year 1) and control (year 2) woredas for households and individuals, respectively, for households found at endline. Balance also holds for eligible and ineligible households separately.

C. *Sample Characteristics*

We designed our geo-referenced household survey to measure key urban outcomes that are rarely available in developing-country cities. In particular, we are able to measure labor market outcomes and commuting flows at the individual level, housing and rents, and local urban amenities.

Employment and Earnings in Addis Ababa: Table 1 compares the labor market outcomes of our sample with outcomes for the city as a whole from a national labor force survey for the year of our baseline survey (Ethiopian Statistical Service 2018). We look at four mutually exclusive and exhaustive categories: employment, available (not working in the last seven days but available for work or engaged in irregular work), in education, and inactive (not in employment or education and unavailable). In our sample of working-age adults in control areas, 48 percent were employed at baseline. The employment rate in the population as a whole is higher, at 56 percent. The relatively low employment rate in our sample is driven in part by high rates of educational enrollment (16.3 percent) and high levels of inactivity (15.0 percent), which includes disability, early retirement, and unpaid work in the home, especially among married women. Unemployment—using the definition of having taken an active step to search for work in the last month—is only 4.7 percent in our sample. It is 6.0 percent in the city as a whole, and this concentrated among the youth. In our sample, among adults over the age of 35, only 1.7 percent are without work and actively seeking work.

Wage employment is the predominant form of employment relative to self-employment: 70 percent of private sector workers in the city are wage employed. Self-employment is even less common in our evaluation sample.[28] We do not observe self-employment earnings in the labor force survey, but a comparison of wage earnings shows that workers in our sample earn roughly half of the representative worker. Wage employment is generally precarious and informal: less than half of wage workers have a permanent contract with job security, and this share is only 21 percent in our evaluation sample.[29] Due to part-time work, employment expressed as a share of 48 weekly hours, our preferred outcome measure of employment, is slightly lower than employment measured at the extensive margin.

Commuting: Our survey data capture individuals' commuting destinations by asking in which woreda they work. Such data are relatively rare in applications of spatial equilibrium models, which usually combine separate wage and commuting data at the neighborhood level. This allows us to do two things in our main estimation: first, it allows us to estimate wages in each destination labor market rather than in each place of residence. Second, we compute commuting flows at the woreda-pair level for baseline and endline, which is essential to estimate how equilibrium effects spill over across woredas. We also ask about commute times, costs, and modes of transport. Table 1 provides basic commuting statistics, and online Appendix C describes the commuting data in more detail. Fifty-five percent of workers commute outside of their neighborhood: on average, they commute over 5 kilometers, for 50 minutes, at a cost of nearly 6 percent of their earnings. The importance of commuting across neighborhoods motivates our analysis of the spillovers of the UPSNP to labor markets across the city.

We use our commuting data to construct a bilateral commuting matrix at baseline and endline. Panels A and B of Figure 2 show out- and in-commuting flows at the woreda level. The woredas that send the most commuters tend to be the central woredas, except a few located at the periphery. Central woredas have higher rates of workers who commute in than those further away, but some peripheral woredas also receive substantial flows of in-commuters.

[28] Among the self-employed, retail (at stalls, markets, and kiosks) makes up roughly 25 percent of self-employment activities in our sample. Other highly represented self-employment activities include tailoring, beauty services, baking, and handicrafts.

[29] We find reasonably high levels of churn: of wage- and self-employed workers at baseline, respectively, 19 percent and 24 percent were either unemployed or inactive one year later. Donovan, Lu, and Schoellman (2020) report global averages of 5.5 percent and 6.9 percent at the *quarterly* interval.

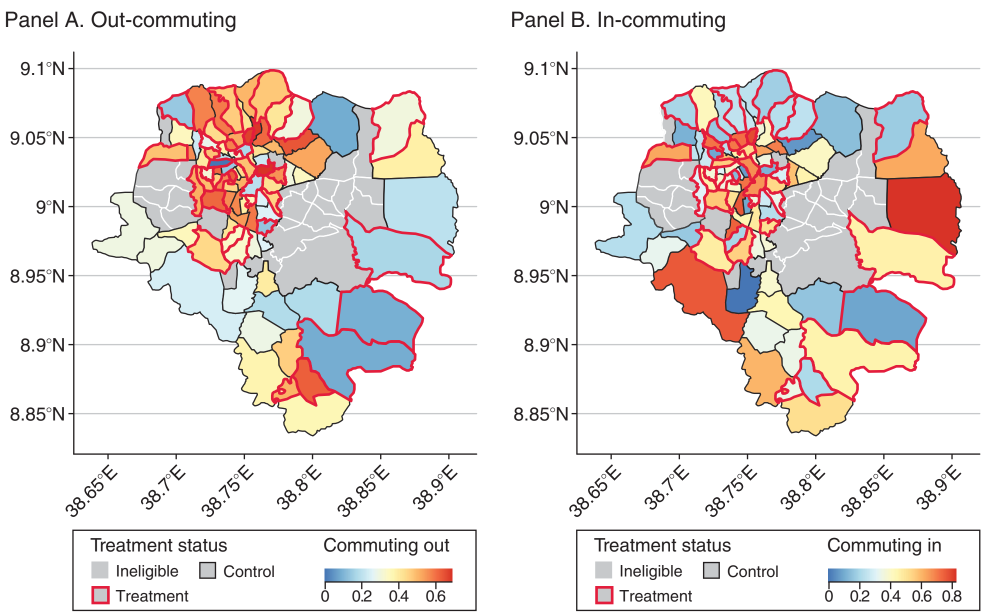

Figure 2. Commuting Rates as a Percentage of Workers by Woreda

Housing and Rents: In our sample, 75 percent of households live in "kebele" housing: this is government-owned housing where households generally live for free or for a nominal fee paid to local government officials. This housing is usually of very low quality; fewer than 10 percent of kebele houses have walls made of formal materials. The average rent for households who do pay rent in this type of housing is Br11 per month, relative to roughly Br660 per month on average in private sector housing. Opportunities to live in kebele housing are rationed, and households cannot move home easily without losing access to these low rents. As a result, mobility rates among households in our sample, and those living in kebele housing in particular, are very low. Only 2.4 percent of our sample moved between the first and second endline survey (over a 21-month period), and only 1.5 percent among those originally in kebele housing.

Amenities: We collected data on neighborhood amenities by asking households to rate the quality of different aspects of their local area. For our main analysis, we use a standardized index comprised of five measures of neighborhood quality, namely, quality of drainage infrastructure, cleanliness of streets, public toilets, presence of odors from sewerage, and presence of odors from trash. This index was prespecified in a pre-analysis plan and was designed to capture improvements to neighborhoods that were likely to result from the activities conducted under the public works. Online Appendix Table A6 summarizes the components for woredas that did not receive the program. Satisfaction with amenities is low. Less than 40 percent of respondents are satisfied with drainage and sewerage systems in their neighborhood; 62 percent of respondents say that they notice the smell of trash in their neighborhood "sometimes" or "very often."

II. Effects on Employment and Amenities

A. *Estimation*

In this section, we estimate the effect of living in a treated woreda T_i on outcome at endline $Y_{\omega hi}$ for worker ω living in household h in woreda i using the following equation:

$$Y_{\omega hi} = \alpha + \beta T_i + \gamma Y_{\omega hi}^0 + \delta \mathbf{X}_{\omega hi} + \varepsilon_{\omega hi}. \tag{1}$$

$Y_{\omega hi}^0$ is the outcome at baseline. The vector $\mathbf{X}_{\omega hi}$ includes baseline individual- and household-level controls and subcity fixed effects.[30] For labor outcomes, we restrict the sample to working-age individuals. Equation (1) can also be estimated at the household level to estimate the treatment effect on any household-level outcome Y_{hi}. This effect is an intention-to-treat (ITT) estimate since some households are eligible for the program and some are not. We observe household eligibility for both treated and untreated woredas: in untreated woredas, these are the households that we observe enrolled in the program in its second year when we conducted an additional endline survey. We also estimate equation (1) separately for eligible and ineligible households.[31] All outcomes and specifications used in this section were prespecified.

B. *Results*

Table 2 presents the program effects on employment (hours worked) and local amenities. Panel A shows the effect of being in a treated woreda where the program is implemented (equation (1)), while panels B and C present separate estimates of equation (1) for eligible and ineligible households, respectively.

The results in columns 1, 2, and 3 suggest that the program generated substantial employment on public works (4.6pp or 12.6 percent of hours worked in the control) but also decreased hours worked in the private sector by 12.8 percent (4.7pp decrease as compared to the control mean of 36.6 percent). In net, there is no change in total hours worked: the coefficient in column 1 is a precisely estimated zero. Panels B and C present the results separately for ineligible and eligible households (eligible households are 44 percent of the sample). As expected, the employment effects are concentrated among eligible households, for whom private sector work declined by 22 percent (8pp as compared to the control mean of 36 percent).

We also show the effects of the program on our index of households' self-reported neighborhood amenities: the program improved the index by about 0.6 standard deviations (column 4, panel A in Table 2). Importantly, this result is not just driven

[30] Individual controls are *gender*, dummies for *age* in bins of ten years, and dummies for education level. Household controls are *gender of household head*; *age of head*; *maximum years of school in the household*; dummies for *household head's educational attainment*; and dummies equal to one if the household *has a disabled member*, *head was employed at baseline*, *rented housing from the government*, *has a formal floor*, *has an improved toilet*.

[31] The estimates of equation (1) could be biased if the effects on labor supply spill over across woredas, which would violate stable unit treatment value assumption (SUTVA). We address this concern in Section IVB after introducing our spatial equilibrium framework and find no evidence of such spillovers.

Table 2—Effects on Employment and Amenities

	Share of hours spent on			
	Employment (1)	Public employment (2)	Private employment (3)	Neighborhood amenities (4)
Panel A. Whole sample				
Treatment	−0.001	0.046	−0.047	0.570
	(0.012)	(0.002)	(0.012)	(0.077)
Control mean	0.366	0	0.366	0
Observations	19,442	19,442	19,442	5,712
Panel B. Eligible households only				
Treatment	0.021	0.101	−0.080	0.603
	(0.015)	(0.003)	(0.014)	(0.083)
Control mean	0.36	0	0.359	−0.007
Observations	8,679	8,679	8,679	2,409
Panel C. Ineligible households only				
Treatment	−0.020	0.001	−0.021	0.543
	(0.013)	(0.0002)	(0.014)	(0.085)
Control mean ineligible	0.378	0	0.378	0.006
Observations	10,763	10,763	10,763	3,303

Notes: In columns 1 to 3, the unit of observation is all working-age adults in our endline sample. In column 4 the sample is composed of one adult respondent per household (the household head or their spouse). "Employment" denotes total hours worked divided by 48 hours per week. "Public employment" denotes hours worked on public works divided by 48 hours per week. "Private employment" denotes hours worked on private sector wage work or self-employment divided by 48 hours per week. "Neighborhood amenities" is a standardized index of answers to five questions about neighborhood quality described in online Appendix Table A6. "Treatment" is a dummy equal to one for households in treated neighborhoods. All specifications include subcity fixed effects, individual and household controls. Standard errors are clustered at the neighborhood level.

by eligible households who directly participated in the work but is present among other residents of the neighborhood who did not do the work (column 4, panel B). Since the program did small-scale neighborhood improvements in beneficiaries' home woredas, these amenity effects are unlikely to spill over to neighboring woredas.

Extensive Margin of Employment: It is surprising that a policy that generates the equivalent of 27 percent of private sector hours worked by eligible households at baseline does not increase employment. In online Appendix Table A9, we show that on the extensive margin the program does increase the employment rate of members of eligible households from 43 to 53 percent (there is no effect among ineligible households).[32] But private sector work participation also declines from 42 to 34 percent among members of eligible households. And since public work hours are

[32] In online Appendix Table A10, we show that two-thirds of this effect come from a decline in inactivity (domestic work and retirement), and one-third from a reduction in the fraction of adults available for work but not working.

typically half of those in the private sector, in terms of hours, the decline in private employment offsets the increase in labor force participation.

Heterogeneity: We provide evidence on effect heterogeneity by type of work and by worker characteristics in online Appendix Table A12. Columns 3 and 4 of online Appendix Table A12 decompose the effect of private employment between self-employment and wage work: both types of work are negatively affected. The reduction in wage work is larger in absolute terms (-3.2pp as compared to -1.5pp for self-employment), but because most people do wage work in the sample, the reduction in hours spent self-employed is larger in relative terms (-18 percent as compared to -11 percent for wage work). Finally, columns 5 to 8 of online Appendix Table A12 present the effects on private employment by gender and skill level. We find that the program reduces private employment for male and female workers, for workers with and without a high school diploma. Consistent with the information on program take-up discussed in Section I, the effects are larger for female and low-skilled workers (especially in proportional terms), but the differences are not statistically significant.

Household-Level Outcomes: Online Appendix Table A8 provides additional results on household outcomes: household income increases, due to public works wages received by eligible households, but household expenditures do not increase; instead, eligible households double their savings. We test whether the improvement in amenities led to an increase in rents in treated woredas or a decrease in the fraction of households moving out of treated neighborhoods. The results in online Appendix Table A7 suggest that rents may have increased by about 3 percent, but the coefficient is not significant due to the small fraction of households who actually pay rents (18 percent). Few households move houses (2 percent), and the proportion is not different in treated woredas. These results are consistent with the fact that poor households in Addis Ababa benefit from government housing and do not pay rent, with little scope for residential mobility (see Section I).

To conclude, the comparison of household outcomes in treated and control neighborhoods suggests that the program improved local amenities but that the employment it generated was entirely offset by a fall in private sector work. Since 18 percent of households in treated areas are in the program, this represents a large negative labor supply shock to the private sector. Also, since about half of the workers in the sample work outside of their neighborhood, this shock likely had spillover effects on labor markets across the city.[33] In the next sections, we use a spatial equilibrium model to quantify the labor market spillovers of the program and combine the direct and indirect effects of the program into a unified welfare analysis.

[33] Online Appendix Table A11 shows that the probability of commuting out decreased by 2.6pp (13 percent) among eligible households in treated as compared to control neighborhoods. This estimate may be biased: since commuters from control neighborhoods work in the same labor markets as those from treated ones, their commuting behavior may also be affected, and SUTVA may not hold.

III. Model

In this section, we model the effects of a public works program in a spatial equilibrium framework of commuting based on Monte, Redding, and Rossi-Hansberg (2018); Heblich, Redding, and Sturm (2020); and Balboni et al. (2021). We consider a city comprising $i = 1, \ldots, n$ locations. In each location i live $\bar{R}_i$ residents, each of whom supplies inelastically one unit of labor. Workers can commute (choose where they work), but they cannot migrate (choose where they live). Let π_{ij} denote the proportion of residents from i who work in j. We assume frictionless trade across the city.

A. *Utility*

Utility for a worker ω residing in location i and working in j is given by

$$U_{ij}(\omega) = B_i \tau_{ij} C_i \epsilon_{ij}(\omega),$$

where C_i denotes consumption of the tradable good, τ_{ij} iceberg commuting costs (≤ 1). B_i is the average amenity from living in i, and $\epsilon_{ij}(\omega)$ is an idiosyncratic amenity shock drawn from a Frechet distribution with dispersion parameter θ:

$$G(\epsilon) = e^{-\epsilon^{-\theta}}.$$

B. *Consumption*

Workers consume a single good, which is freely traded across the city. We use its price as numeraire. Utility maximization implies that workers consume all of their income on goods. Let $\bar{v}_i$ denote the average income of workers living in i and C_i denote aggregate consumption:

$$C_i = \bar{v}_i.$$

C. *Production*

We assume that production in each location is made by a representative firm that uses only labor as production factor:

$$Y_j = A_j F(L_j), \quad \text{where} \quad F'(\cdot) > 0 \quad \text{and} \quad F''(\cdot) < 0.$$

The productivity term A_j is fixed. All firms produce the same product, whose price is one. We denote the labor demand elasticity with ε_D and show in online Appendix B.2 that it is negative:

$$\frac{\partial \ln L_j}{\partial \ln w_j} = \varepsilon_D < 0.$$

D. *Commuting*

Utility is linear, and the budget constraint imposes $C_{ij} = w_j$; hence, the utility from living in i and working in j is

$$U_{ij} = B_i \epsilon_{ij} \tau_{ij} w_j.$$

Because there is no mobility, utility is not necessarily equalized across locations of residence. However, it is still equal within a location of residence across the different possible destinations. We show in online Appendix B.1 that the expected utility of a location of residence i is

$$(2) \qquad \forall i,\ U_i = \gamma \left[\sum_{j=1}^{n} \left(B_i \tau_{ij} w_j \right)^{\theta} \right]^{\frac{1}{\theta}}, \quad \text{where} \quad \gamma = \Gamma\left(\frac{\theta - 1}{\theta} \right).$$

We show in online Appendix B.3 that the probability that a worker who lives in i will work in j is

$$(3) \quad \pi_{ij} = \frac{\left(B_i \tau_{ij} w_j \right)^{\theta}}{\sum_k \left(B_i \tau_{ik} w_k \right)^{\theta}} = \frac{\Phi_{ij}}{\Phi_i}, \quad \text{where} \quad \Phi_{ij} = \left(B_i \tau_{ij} w_j \right)^{\theta}, \quad \Phi_i = \sum_j \Phi_{ij}.$$

E. *General Equilibrium*

Given the endowments A_i, B_i, and R_i, the commuting costs τ_{ij}, the parameter θ, and $F(\cdot)$, an equilibrium is a vector of wages w_j in each location, which ensures that the labor markets clear:

$$(4) \qquad \forall j,\ L_j = \sum_i \pi_{ij} R_i.$$

Monte, Redding, and Rossi-Hansberg (2018) show that this equilibrium exists and is unique. We will use the "exact hat" algebra, popular in trade (e.g., Arkolakis, Costinot, and Rodríguez-Clare 2012) to study the effect of the program on this equilibrium. For every variable X, X denotes the equilibrium value without the program, X' the equilibrium value with the program, and $\hat{X} = X'/X$ the effect of the program on X.

F. *Public Works*

Let T_i be the treatment indicator equal to one if the public works program is implemented in location i. If $T_i = 1$, the program offers to workers who live in i the opportunity to work locally (without commuting costs) for p part of their time at a wage w_g. Each worker ω receives a Frechet-distributed idiosyncratic utility shock ϵ_g from working in the program. We assume that there is full take-up of the program, so that in treated areas, labor supply to the private sector is $1 - p$. The commuting probabilities π_{ij} are now defined conditional on doing private sector work.

The program has three effects. First, it brings a net direct income gain equal to public works wages minus forgone income from the private sector, multiplied by the share of labor supply dedicated to public works p in treated locations:

$$\text{(5)} \qquad \textit{Direct Income Gain} = pT_i\Big(w_g - \sum_j \pi_{ij} w_j\Big).$$

Second, the program changes the labor market equilibrium: workers who participate in public works in treated locations reduce their labor supply to the private sector in each commuting destination. Given the expression of the labor demand elasticity, we show in online Appendix B.4 that effect of the program on the wage in each location j can be written as

$$\text{(6)} \qquad \ln \hat{w}_j = \frac{1}{\varepsilon_D} \ln\Big[\sum_i \lambda_{ij} \hat{\pi}_{ij}(1 - pT_i)\Big],$$

where λ_{ij} is the fraction of people who work in j that comes from i in the no-program equilibrium, which determines the exposure of labor market j to labor supply shocks from i. Wages will rise in each labor market proportionally to the fraction of the workforce that comes from treated locations. Since labor markets are integrated across the city, this fraction will be higher in the treated locations themselves but will not be zero in places that do not get the program. The expression includes the immediate reduction in labor supply to the private sector in treated areas $(1 - pT_i)$ but also an endogenous response of commuting probabilities for private sector workers $\hat{\pi}_{ij}$, which will dampen the effects of the negative labor supply shock on wages.

Third, the program improves local amenities for all residents of treated locations. Let $\hat{B}_i$ denote the relative change in amenities:

$$\hat{B}_i = (1 + bT_i).$$

We show in online Appendix B.5 that the expected utility for a worker living in i when the program is implemented writes

$$\text{(7)} \qquad U_i' = \gamma(1 + bT_i)\left\{pT_i B_i w_g + (1 - pT_i)\Big[\sum_j \hat{w}_j^{\theta}(B_i \tau_{ij} w_j)^{\theta}\Big]^{\frac{1}{\theta}}\right\}.$$

G. *Welfare Effects*

Based on equations (2) and (7), we can derive the welfare gains from the public works program (see proof in online Appendix B.6). We have rearranged terms to provide the following decomposition:

$$\text{(8)} \qquad \hat{U}_i = \underbrace{(1 + bT_i)}_{\textit{Amenity Effect}}\left\{1 + \underbrace{pT_i\Big[(1 + g_i)\pi_{ii}^{\frac{1}{\theta}} - 1\Big]}_{\textit{Direct Effect}} + \underbrace{(1 - pT_i)\Big[\Big(\sum_j \pi_{ij}\hat{w}_j^{\theta}\Big)^{\frac{1}{\theta}} - 1\Big]}_{\textit{Wage Effect}}\right\},$$

which includes the effect of improved amenities, the direct gains from participation in the program (including the reduction in earnings due to reduced labor supply to private sector holding wages at their nonprogram level), and the gains from rising private sector wages. It can be computed with the knowledge of p (share of the labor supply devoted to the program and taken away from private sector work), $(1 + g_i) = w_g/w_i$ the woreda-specific wage premium on public works, $\hat{w}_j$ the effect of the program on private sector wages, π_{ij} the commuting probabilities without the program, θ the Frechet parameter, and $(1 + b)$ the proportional change in the value of local amenities.

As a benchmark, we will compare the welfare gains from the program with the benefits from a cash transfer that provides the same utility as public works wages without any work requirement and hence has no labor market effect (see online Appendix B.7 for more details and Section IVE for further discussion):

$$(9) \qquad \hat{U}_i^{cash} = 1 + pT_i(1 + g_i)\pi_{ii}^{\frac{1}{\theta}}.$$

H. *Discussion*

The model abstracts from a few dimensions that may be important in other contexts: housing, trade, migration, capital, labor supply adjustments, and taxes.

The absence of housing markets in the model is motivated by a context in which poor households receive housing from the government, rarely pay rents, and rarely change residence. The model does not consider the goods market either and potential effects on local prices. This is motivated by the fact that goods markets within a city are likely to be well integrated. These assumptions are also supported by our reduced-form estimates: as discussed in Section II, we find small and insignificant effects on rents and residential mobility (see online Appendix Table A7) and no evidence that the program increased household expenditures (online Appendix Table A8).[34] Our setting in this regard differs both from studies of urban transportation programs, which document gentrification and rising rents (Tsivanidis 2019; Balboni et al. 2021), and from studies of rural social protection programs, which document large increases in consumption and rising prices in more remote areas (Cunha, Giorgi, and Jayachandran 2019; Egger et al. 2022).

We also abstract from three mechanisms that could dampen the program effect on wages. First, we allow changes in commuting to shift labor supply to different labor markets but assume that total labor supply from each location is fixed. This assumption is justified by the fact that total hours worked are not affected by the program, with a one-for-one crowd-out of private sector work by public works (Table 2). Second, we ignore migration into the city, like most papers in the literature on urban spatial equilibrium do (Tsivanidis 2019; Balboni et al. 2021). Residential eligibility criteria make it impossible for migrant households to move to the city in order to enroll in the program, but they may move in order to benefit from higher private sector wages. This migration response could in the longer run dampen the program

[34] We use official geo-referenced micro data from the Consumer Price Index to test empirically whether the program had any effect on local prices and do not find evidence of price effects (see online Appendix D and online Appendix Table D2).

effect on wages and reduce the welfare gains to residents but also, presumably, increase the welfare of migrants. A third mechanism that would mitigate the effect of the program on wages would be a downward adjustment in labor demand, e.g., by substitution of capital for labor. These forces are more likely to play in the longer run, which makes them less relevant for our study since the UPSNP is implemented for only a few years.

Finally, the funding of the program is outside of the model: the UPSNP is funded by a World Bank loan, to be repaid by the federal government, whose main sources of revenue are corporation tax, income tax, trade tax, and VAT, which have low incidence on poor households (Harris and Seid 2021).

IV. Quantitative Analysis

A. *Effect on Private Sector Wages*

An estimation of the wage effects of the program that does not consider commuting and spatial spillovers would follow the approach from Section II and compare wages earned by residents of treated woredas with wages earned by residents of control woredas. In this section, we use woredas as the unit of analysis to correspond with locations i and j in the model. We will refer to the woreda in which someone lives as their *neighborhood* and the woreda in which they work as their *labor market*. Following the model notation, let us denote with T_i the treatment dummy for neighborhood i, and with w_i the average wage earned by workers who live in i. The specification from Section II is

$$\ln w_i \;=\; \alpha + \beta T_i + \gamma \ln w_i^0 + \delta \mathbf{X}_i + \varepsilon_i, \tag{10}$$

where $\ln w_i^0$ are baseline wages and $\mathbf{X}_i$ a vector of average worker characteristics (gender, age, and education dummies), as well as subcity fixed effects. In order for this specification to provide unbiased estimates of the effect of the program, the stable unit treatment value assumption (SUTVA) needs to hold; i.e., wages in a given neighborhood should not be affected by the implementation of the program in other neighborhoods. Given the importance of commuting flows across woredas, SUTVA is unlikely to hold. In particular, equation (6) in the model makes it clear that the wage effects of the program are better captured by wages at place of work, rather than place of residence, and are proportional to changes in labor supply of commuters coming from treated neighborhoods.

Taking equation (6) to the data is challenging because the labor supply change includes an exogenous reduction due to the program and an endogenous change in commuting patterns in response to wage changes. There are two possible approaches to this issue: a reduced-form approach, which regresses wages on the exogenous component only, and a more structural approach, which regresses wages on labor supply instrumented by the exogenous component.[35] We first adopt the reduced-form approach and discuss the instrumental variable findings below. We

[35] In a similar vein, Donaldson and Hornbeck (2016) adopt a reduced-form approach to estimate the effect of market access.

set $\hat{\pi}_{ij} = 1$; i.e., we ignore the endogenous change in commuting probabilities from i to j due to wage changes in j. With a linear approximation of the log function, we obtain (see online Appendix B.4 for more details)

$$\ln \hat{w}_j = \frac{1}{\varepsilon_D} \ln\left[\sum_i \lambda_{ij} \hat{\pi}_{ij}(1 - pT_i)\right] \approx -\frac{p}{\varepsilon_D} \sum_i \lambda_{ij} T_i. \tag{11}$$

We can then take equation (11) to the data and consider as an outcome private sector wages earned by workers who work in a labor market j, which we can construct thanks to the commuting data at the individual level. We regress wages on exposure to the program in that labor market:

$$\ln w_j = \alpha + \beta Exposure_j + \gamma \ln w_j^0 + \delta \mathbf{X}_j + \varepsilon_j, \tag{12}$$

where $\ln w_j^0$ are baseline wages in labor market j and $\mathbf{X}_j$ is a vector of characteristics of workers who work in j (age, gender, and education dummies).[36] Exposure to the program is defined as

$$Exposure_j = \left(\sum_i \lambda_{ij} T_i - \frac{1}{R} \sum_{0 \le r \le R} \sum_i \lambda_{ij} \tilde{T}_i^r\right),$$

where T_i is a dummy for the implementation of the program in neighborhood of residence i and λ_{ij} is the probability at baseline that a worker who works in neighborhood j lives in neighborhood i. Our approach is similar to a shift-share instrument as in the migration literature (Imbert et al. 2022): β causally identifies the effect of exposure to the program. Note that $i = j$ is one of the elements of the sum, so that the coefficient β captures the effect of the program on local wages as well as its effect on wages in other neighborhoods. Our setting is similar to the one of Borusyak and Hull (2023): neighborhoods are nonrandomly exposed (through commuting shares) to a randomly allocated shock (the program). To avoid an omitted variable bias, we follow Borusyak and Hull (2023) and re-center actual exposure using average potential exposure from 2,000 simulated independent treatment assignments $\tilde{T}_i^r$ that follow the same (stratified) random allocation.

To give a graphical illustration of our argument, Figure 3 plots log wages in each labor market as a function of (noncentered) exposure, for treated and control labor markets separately. Because 55 percent of workers live and work in the same neighborhood, treated labor markets are the most exposed to the change in labor supply due to the program. However, control labor markets are also exposed to some extent, and there is heterogeneity in exposure within the two groups. The figure makes it clear that log wages are linearly increasing as a function of exposure to the program, with approximately the same slope among treated and untreated labor markets.

Table 3 presents the estimates of β based on the two specifications (10) and (12). In column 1, the comparison between control and treated neighborhoods from equation (10) suggests that wages earned by workers living in treated neighborhoods

[36] Notice that $\ln \hat{w}_j$ in the model indicates differences in wages between equilibria induced by the program, not differences between time periods at the woreda level. The intercept α in equation (12) now identifies the counterfactual wage in a woreda with zero exposure. We control for $\ln w_j^0$ for increased efficiency; this is also efficient relative to use of changes between time periods as the outcome variable (McKenzie 2012). Our results are similar without this control.

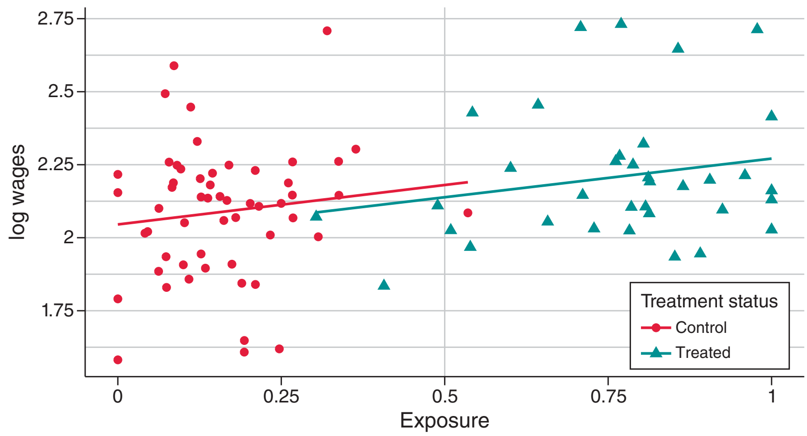

Figure 3. Private Sector Wages as a Function of Exposure to the Public Works Program in Treated and Control Neighborhoods

Note: Estimated slope coefficients are for "Control" 0.3489 (0.3026) and for "Treated" 0.2606 (0.1940).

increased by 10.2 percent. Column 2 presents the same results controlling for worker characteristics, and the coefficient drops slightly to 9.3 percent. In column 3, the model-based estimate suggests that a labor market that would draw all its labor supply from treated areas would see its wages increase by 21.4 percent. Once we control for worker characteristics, the coefficient drops slightly to 18.6 percent, our preferred estimate. The decline in coefficients after including controls suggests positive selection of private sector workers: as Section IA shows, women and less educated workers, who earn lower wages, are more likely to participate in public works and hence more likely to exit the private sector. But these composition effects are small, and our estimate points to large equilibrium effects.

On average, treated labor markets receive 77 percent of their labor supply from treated neighborhoods, against 16 percent for control labor markets. Our preferred estimate of 18.6 percent implies that in the partial rollout of the program, wages have increased by 14 percent in treated labor markets and 3 percent in control labor markets. By contrast, estimates based on equation (10) would have implied a 9.3 percent rise for residents of treated woredas and no effect in control. Hence, the estimates from equation (10) that ignore commuting and labor market spillovers miss a sizable rise in wages in control woredas and severely underestimate the rise in wages in treated woredas. What is more, our preferred estimate implies that once fully rolled out, the program will increase wages by 18.6 percent in all woredas and not 9.3 percent as the simple treatment versus control comparison would suggest.[37]

Robustness: In columns 1 and 2 of online Appendix Table A14, we address the concern that our survey-based measure of commuting probabilities λ_{ij} between 90 origins and 90 destinations may be too sparse (Dingel and Tintelnot 2020). For this, we perform two robustness checks: (i) we predict commuting probabilities based

[37] Under full rollout, $T_i = 1$ everywhere; hence, $Exposure_j = 1$ everywhere.

Table 3—Effect of the Public Works Program on Private Sector Wages

	log wages at origin		log wages at destination	
	(1)	(2)	(3)	(4)
Treatment at origin	0.102	0.093		
	(0.037)	(0.040)		
Exposure of destination			0.214	0.186
			(0.074)	(0.074)
RI p-values	0.0235	0.007	0.0005	0.013
Worker controls	No	Yes	No	Yes
Observations	90	90	90	90

Notes: The unit of observation is a neighborhood. In columns 1 and 2, the dependent variable is log hourly earnings from private sector work (excluding public works) earned by workers who live in that neighborhood. In column 1 the specification includes only subcity fixed effects. In column 2 the specification also includes worker controls. In columns 3 and 4, the dependent variable is log hourly earnings from private sector work (excluding public works) earned by workers who work in that neighborhood. In column 3 the specification does not include any controls. In column 4 the specification controls for the characteristics of workers who work in the neighborhood. "Treatment" is a dummy equal to one if the neighborhood is treated. "Exposure" of a neighborhood j is defined as the sum of the treatment status of each neighborhood i weighted by the fraction of residents from i who work in neighborhood j. The sum includes neighborhood j itself. Exposure is re-centered following Borusyak and Hull (2023) using average exposure across 2,000 simulated treatment assignments. RI p-values are p-values obtained through randomization inference, with 2,000 simulated treatment assignments.

on a Poisson model, or alternatively, (ii) we infer the commuting probabilities of respondents who did not report where they worked based on the commuting probabilities of respondents who did. The estimates based on these alternative measures of exposure are similar to our main estimate.[38]

Labor Demand Elasticity: Equation (11) identifies the effect of the exogenous component of a labor supply induced by the program and therefore causally identifies the effect of exposure to the program. However, it does not identify the labor demand elasticity since we obtained equation (11) (i) by shutting down endogenous changes in commuting probabilities ($\hat{\pi}_{ij} = 1$) and (ii) by approximating $\ln\left[1 - \sum_i p\lambda_{ij}T_i\right]$ with $-p\sum_i \lambda_{ij}T_i$ (see proof in online Appendix B.4). We undo the approximation first: column 3 of online Appendix Table A14 regresses log wages on the log form of exposure $\ln\left[1 - \sum_i p\lambda_{ij}T_i\right]$ (setting p to 0.128 based on the ITT estimates). The implied labor demand elasticity is the inverse of the coefficient $\varepsilon_D = -0.74$. This estimate of labor demand elasticity may still be biased in the presence of endogenous changes to commuting probabilities $\hat{\pi}_{ij}$. In column 4 we estimate the structural equation: we regress log wages at destination on log labor supply to that destination (including changes in commuting probabilities)

[38] As an additional robustness check, online Appendix Table A15 presents the results on daily wages rather than hourly wages and finds similar effects.

instrumented by the log form of exposure. The implied labor demand elasticity is the inverse of the coefficient $\varepsilon_D = -0.68$.

Heterogeneity: Table A13 in the online Appendix provides heterogeneity analysis. We estimate the labor market spillovers separately for eligible and ineligible households and show that the wage effects are also felt by ineligible households (columns 1 and 2). Since ineligible households do not reduce their labor supply to the private sector (panel C in Table 2), this result provides further evidence that composition effects are not driving the increase in wages. We also check that labor market spillovers are present in both self-employment and wage work (columns 3 and 4 of online Appendix Table A13). The effects are stronger (but not significantly so) for the 25 percent of workers who are self-employed and who also experienced a larger reduction in labor supply (see online Appendix Table A12). These results suggest that our modeling choice to assimilate self-employed to wage workers, with only one firm per location, is a reasonable approximation: the main difference is that in reality the labor supply of self-employed workers shifts along their own labor demand curve rather than their employers'. In columns 5 and 6 of online Appendix Table A13, we estimate separately the effect of the program on wages of male and female workers: although female labor supply declined more than male labor supply, the wage effects are similar for men and women, and if anything higher for men, but the coefficients are not statistically different. Finally, we also find that the effects are concentrated among low-skilled workers, with a more muted and statistically insignificant effect for skilled workers (columns 7 and 8), but the difference between the two coefficients is once again not significant.

Alternative Approaches: Our approach relies on a model-based measure of exposure through the commuting network, but we also test whether more commonly used methods based on distance to treated units reach similar conclusions. In online Appendix E, we consider two methods: the "donut" approach, which drops from the analysis untreated units that are close to treated units, and the "radius" approach, which compares untreated units within a certain radius of treated units with those further away. The estimates range from large negative to large positive values, they are very sensitive to the choice of the radius over which exposure is defined, and the standard errors are very large throughout, so that no estimate is significant. Hence, the methods commonly used in the literature do not adequately capture spillovers in our setting.

B. *Employment Spillovers*

A natural question is whether wage spillovers induced employment spillovers, i.e., whether vacancies left by workers who participated in the program were filled by other workers. In Section II we found no evidence of within-woreda spillovers: ineligible households in treatment woredas did not work more than ineligible households in control woredas (panel C of Table 2). There could still, however, be spillovers across woredas, e.g., if workers in control woredas increased their labor supply to fill in for workers in treated woredas. Importantly, these spillovers would violate SUTVA and bias our estimates of the employment effects of the program, which we obtained by simply comparing employment outcomes between residents of treatment and control woredas (Table 2).

To test this, we construct *Exposure Squared*, which captures second-order effects of the program. *Exposure Squared* measures the extent to which residents of a neighborhood i may be affected by changes in wages in each labor market j due to its exposure to changes in labor supply induced by program implementation in each neighborhood k. Formally,

$$ExposureSquared_i = \sum_j \pi_{ij} Exposure_j = \sum_j \pi_{ij}\left(\sum_k \lambda_{jk} T_k - \frac{1}{R}\sum_{0 \leq r \leq R} \sum_k \lambda_{jk} \tilde{T}_k^r\right),$$

where π_{ij} is the baseline probability of commuting from i to j and λ_{jk} is the share of workers in k who come from j. We regress employment outcomes of a worker who lives in i on whether i is treated and on i's *ExposureSquared*. Online Appendix Table F1 present the results: we find similar employment effects from living in a treated woreda and no evidence of employment spillovers. We refer the interested reader to online Appendix F for more detail.

The large wage effects and the absence of employment spillovers from the UPSNP are the mirror image of Breza, Kaur, and Shamdasani's (2021) experimental evidence of surplus labor in rural India. They hire part of the workforce in randomly chosen villages in the lean season and find no effect on wages or on employment because other workers pick up the slack. Our results point to the opposite conclusion to theirs, i.e., that urban Ethiopia is not characterized by surplus labor. They also confirm that in the context we study, household labor supply is inelastic.

C. *Effect on Local Amenities*

The ITT results in Section II suggest that the public works program improved local amenities in the neighborhoods where it is implemented. Specifically, we measure amenities through a standardized index of qualitative assessments by residents on different dimensions of neighborhood quality and show that the index increases by 0.570 in neighborhoods with the program.[39] To estimate the welfare gains from better amenities, we need to convert the increase in index quality into a monetary equivalent. If housing markets were fully functional, one would expect this increase in amenities to be reflected in increase in rents paid by households. However, as we discussed in Section II, only 18 percent of poor households in this context pay rents, so that the program effect on rents paid is a positive but insignificant 0.033 (online Appendix Table A7).

To provide a more precise valuation of the improvement of amenities, we combine information on rents paid by households who do pay rent and information on hypothetical rents for those who do not, i.e., on the value that households think they could expect to pay if they were renting the place they live in, and we compute the correlation between these rents and the neighborhood quality index.[40] Column 1 in

[39] Since the public works make small-scale improvements to local amenities, we assume that they do not spill over to nontreated neighborhoods. We use our exposure-based measure to test for spillovers via the commuting network and find no evidence of spillovers for amenities (results not reported here).

[40] For this exercise, we use our second endline data, for which we have these hypothetical rent data (we do not ask this question to everyone in the first endline) as well as amenities data.

Table 4—Correlation between Neighborhood Quality and Hypothetical Rents

	log hypothetical rent	
	(1)	(2)
Neighborhood quality	0.046	0.044
	(0.017)	(0.013)
Controls	No	Yes
Observations	4,753	4,753

Notes: The unit of observation is a household, interviewed in our second endline. The dependent variable is the log of rents that each household pays for its housing or how much it would pay if it were to rent it (for households who own their housing or do not pay rents). The neighborhood quality index is a standardized index of answers to five questions about neighborhood quality described in online Appendix Table A6. In column 2 the specification includes household and housing controls selected by double lasso. Standard errors are clustered at the neighborhood level.

Table 4 presents the raw correlation between index quality and log rents, which is 0.046. One might worry that household or housing characteristics may be correlated both with neighborhood quality and rents (e.g., household income or housing size). To alleviate this concern, we implement a double postselection lasso procedure to select within a long list of household and housing characteristics those that are the best predictors of either neighborhood quality or rents and include them in the regression. The correlation coefficient, shown in Table 4, column 2, remains very similar after including these controls (0.044). We combine this coefficient and the increase in the index to compute the improvement in amenities due to the public works in monetary terms: $0.574 \times 0.044 = 0.025$, which is in the same ballpark as the imprecisely estimated 3.5 percent increase in rents.

D. *Commuting Elasticities*

To estimate the key model parameter θ, we derive a gravity equation from the expression of the commuting probabilities (equation (3)):

$$\ln \pi_{ij} = \theta \ln w_j + \theta \ln B_i + \theta \ln \tau_{ij} + \Phi_i,$$

where $\Phi_i = \sum_k \left(B_i \tau_{ik} w_k\right)^\theta$ is fixed at the residence level. We substitute iceberg commuting cost $\tau_{ij} < 1$ with commuting costs $c_{ij} = 1/\tau_{ij} > 1$ to obtain

$$\ln \pi_{ij} = \theta \ln w_j + \theta \ln B_i - \theta \ln c_{ij} + \Phi_i.$$

We estimate θ as the elasticity of commuting with respect to wages with the following Poisson specification:

$$\pi_{ij} = \exp\!\left(\theta \ln w_j - \theta \ln c_{ij} + \nu_i + \varepsilon_{ij}\right),$$

where π_{ij} is the share of private sector workers who reside in i who commute to a destination j at endline, $\ln w_j$ is the log of the wage at destination, $\ln c_{ij}$ is our survey measure of the monetary cost of commuting from i to j, and ν_i is a residence fixed

Table 5—Commuting Elasticity with Respect to Wages

	Commuting probability		log destination wage
	Poisson	*Poisson-IV*	First-stage *OLS*
	(1)	(2)	(3)
log destination wage	0.499	2.077	
	(0.299)	(1.236)	
Destination exposure to program			0.188
			(0.074)
log walking time	−2.106	−2.127	0.014
	(0.103)	(0.103)	(0.031)
Observations	7,744	7,744	7,744

Notes: The unit of observation is a neighborhood origin × destination pair. The dependent variable is the commuting probability. "log destination wage" is the log of private sector income per hour earned by workers who work in the neighborhood of destination. "Destination exposure to the program" is for each neighborhood of destination j equal to the sum of treatment status of all neighborhoods i weighted by the commuting probability from i to j. Following Borusyak and Hull (2023), we re-center actual exposure using average exposure to 2,000 simulated treatment assignments. "log walking time" is the log of minutes needed to walk between the centroid of the origin and destination neighborhoods according to Google API. In column 1 the estimation is done with OLS. In column 2 log destination wage is instrumented with the destination exposure to the program. Column 3 presents the first stage of the estimation. All specifications include origin fixed effects. Standard errors are clustered at the destination level.

effect that captures residential amenities in i and average expected utility of workers who live in i. This equation allows us to estimate θ, but only if we can deal with the endogeneity of the wage response to changes in commuting, which in the model is described by equation (6). We use exposure to the program as instrument for the wage w_j.[41]

Table 5 presents the results. Column 1 presents the OLS estimate for the correlation between the wage at destination and commuting probability. The correlation is positive, which is expected given that commuters are more likely to go to a destination with higher wages. This estimate is, however, likely to be downward biased because more commuting will decrease wages at destination. The IV estimate presented in column 2 is much larger in magnitude and significant and implies that the Frechet parameter $\theta = 2.08$. Ours is comparable to recent estimates of the Frechet parameter in other developing-country cities: Tsivanidis (2019) for Bogotà, and Kreindler and Miyauchi (2023) for Dhaka and Colombo. The first stage presented in column 3 is almost identical to the estimation of equation (12), confirming the positive relationship between exposure and destination wages. In online Appendix G, we propose an alternative strategy inspired by Heblich, Redding, and Sturm (2020), who estimate θ as the elasticity of commuting to commuting costs. We find a higher estimate $\theta = 4.05$, which is consistent with Heblich, Redding, and Sturm's (2020) own findings. We use $\theta = 2.08$ to quantify the welfare effects in the next section but present results with $\theta = 4.05$ in online Appendix G.

[41] There is no mechanical effect of local public works participation on commuting probabilities in treated areas because these probabilities are conditional on private sector work.

E. *Welfare Effects for the Urban Poor*

Finally, we combine reduced-form and structural estimates to compute the welfare effects of the program for the representative poor resident of neighborhood i, based on equation (13) from the model:

$$\hat{U}_i = \underbrace{(1 + bT_i)}_{\text{Amenity Effect}} \left\{ 1 + pT_i \left[(1 + g_i) \pi_{ii}^{\frac{1}{\theta}} - 1 \right] + (1 - pT_i) \left[\left(\sum_j \pi_{ij} \hat{w}_j^{\theta} \right)^{\frac{1}{\theta}} - 1 \right] \right\}, \tag{13}$$

where π_{ij} are commuting probabilities, which vary across neighborhoods. Based on Table 2, the fraction of the labor supply devoted to public works is almost equal to the fraction taken away from the private sector $p = 4.7/36.6 = 12.8pp$. The equation includes improvement in amenities by the program, which we have valued at $b = 2.5\%$. It also includes the wage premium g_i, which is the ratio between average hourly earnings on public works and in the private sector for each neighborhood (the average wage premium across all woredas is 1.67). Finally, it includes the changes in wages due to the program, which at the beginning of this section we estimated to be $\hat{w}_j = 0.186 \sum_i \lambda_{ij} T_i$ at destination labor markets j. Residents of neighborhoods i then benefit from these wage changes across all labor markets j through the commuting network π_{ij}. These gains are mediated by the Frechet parameter θ, which we have estimated to be $\theta = 2.08$.

We do this first in the context of the partial rollout of the program and estimate separately the welfare effects for areas with and without the program, and then in the context of the complete rollout of the program, in which all neighborhoods are treated and, therefore, $\hat{w}_j = 0.186$ everywhere. We also sequentially add the different parts of the welfare effects to show their contribution: first the direct gains from participation in public works, then wage spillover effects, then the improvement in amenity. Table 6 presents the results. In the partial rollout, treated neighborhoods experience large welfare gains (16.2 percent), including 3.2 percent from direct gains from participation and 10.2 percent from rising private sector wages. By contrast, control neighborhoods experience a 4.4 percent increase in welfare, which is entirely due to substantial labor market spillovers: the welfare gains from wage increases in untreated areas are 44 percent those of treated areas. We next estimate welfare gains to the poor in the complete rollout scenario. The welfare gains are larger (22.5 percent) overall, an increase that is driven by stronger labor market spillover effects (16.2 percent), while the direct benefits and the amenity effects are basically unchanged. These results make it clear that labor market spillovers are an important part of the welfare effects of the program.

As a benchmark, we estimate the welfare gains from a counterfactual policy, a cash transfer that would provide to households the utility equivalent of wages received on public works. As compared to public works, this hypothetical cash transfer has the advantage of not imposing any work requirements, so that labor supply to the private

TABLE 6—WELFARE EFFECTS OF THE PUBLIC WORKS PROGRAM FOR THE URBAN POOR

Rollout:	Partial		Complete
	Control (1)	Treatment (2)	All (3)
Treatment	0.000	1.000	1.000
Exposure	0.161	0.765	1.000
Direct effect	0.000	0.032	0.033
Direct + wage effects	0.044	0.134	0.195
Direct + wage + amenity	0.044	0.162	0.225
Cash transfer	0.000	0.160	0.162

Notes: Column 1 reports welfare gains to the poor from the public works program in untreated areas under partial rollout. Column 2 reports welfare gains in treated areas under partial rollout. Column 3 reports welfare gains when the program is implemented everywhere. "Exposure" for a given labor market j is equal to the sum of treatment status of all neighborhoods i weighted by the commuting probability from i to j. Rows 3 to 6 show welfare effects for the representative resident of neighborhood i. "Direct effect" is the welfare benefits from participating in the program. "Direct + wage effects" is the sum of the direct effect and the effect of rising private sector wages due to labor market spillovers. "Direct + wage + amenity" is the sum of the direct, the wage effect, and the welfare gains from improved amenities. "Cash transfer" is the welfare gain from a cash transfer program that would give the same utility as participation in the public works without crowd-out of private sector employment.

sector is unchanged.[42] At the same time, because labor supply is unaffected, there are no equilibrium wage effects. To the extent that such a program may have stimulus effects through the goods market as in Egger et al. (2022), the stimulus effect of workfare wages should be similar. As it happens, we find no evidence for such effects, which may be due to our urban setting where households have better access to savings and prices are more likely to be determined outside of the local economy. As the results in Table 6 show, the cash transfer does better than public works only if one focuses on the direct gains from participation. Once indirect effects on amenities and private sector wages are taken into account, the conclusion is overturned, and public works dominate cash. Online Appendix Table A16 provides confidence intervals for the welfare estimates by bootstrapping each parameter in (13) using its estimated standard error. We also bound the difference between the welfare gains from the public works program and the hypothetical cash transfer: the difference is significant at the 5 percent level. Online Appendix G also shows that these conclusions are robust to using an alternative estimate of $\theta = 4.05$.

In online Appendix H, we develop a quantification of the income gains from the program that does not rely on any modeling assumption about utility but ignores the gains from improved amenities. The results are very similar: the wage effects are more than two times larger than the direct effects, and taking them into account tips the balance in favor of public works against a cash transfer that would pay the equivalent of public works wages without any work requirement. We also compute income gains we would have predicted if we had ignored wage spillovers across neighborhoods and estimated wage effects by comparing treated and untreated areas as in Table 3, column 2. We find that we would have underestimated the income

[42] The literature on cash transfers in developing countries suggests that their effects on poor households' labor supply are negligible (Banerjee et al. 2017).

gains under full program rollout by about a third and incorrectly concluded that the cash transfer delivered higher income gains.

F. *Discussion*

Our welfare analysis quantifies the effects of the program for the urban poor in neighborhoods that were eligible in the first year of the program, for which our sample is representative, and not for the entire population of Addis Ababa. In particular, it does not speak to the effect of the program on about 14 percent of the poor in Addis Ababa who do not live in year 1 eligible neighborhoods. They may gain indirectly from the rise in wages we document to the extent that they work in the same labor markets as poor people in eligible neighborhoods. These neighborhoods did receive the program in year 2 and might plausibly have experienced direct benefits like the other neighborhoods. There may also be indirect effects of the program on households who are too rich to be targeted by the policy, in both eligible and ineligible neighborhoods. On the one hand, richer households who live in the same neighborhoods as poor households may gain from improved amenities, and those who work in the same labor market as poor workers will benefit from rising wages. On the other hand, some of the richer households will be employers and will suffer welfare losses if they have to pay higher wages. These distributional effects through the labor market have been highlighted by the literature on rural public works (Imbert and Papp 2020; Muralidharan, Niehaus, and Sukhtankar 2023). Richer households may also pay higher taxes when the federal government repays the World Bank loan that funded the UPSNP.

The magnitude of the direct benefits from the program (program earnings minus forgone private earnings) as well as its indirect effects (through the rise in private sector wages) crucially depend on the extent to which the program crowds out labor supply to the private sector. In our study we find that for every hour provided on public works, there is one fewer hour worked in the private sector, so that the direct benefits are small, but the equilibrium effects large. In contexts where public works do not have large crowd-out effects on private employment, e.g., the public works program in Côte d'Ivoire, which targets young unemployed workers (Bertrand et al. 2021), one would anticipate larger direct benefits and smaller equilibrium effects.

Another important question is what the effects of the program would be if it were continued beyond its planned life-span of three years and became a permanent program similar to the Indian rural employment guarantee, MG-NREGS. As discussed in Section III, if more people move into treated areas in the longer run, poor households who depend on the private housing market may see rents increase, and the welfare gains from better amenities may be partly captured by richer households and landlords (Balboni et al. 2021). Similarly, part of the wage gains may dissipate if labor supply increases as more poor people move into the city or if labor demand decreases with employers substituting capital for labor (Imbert et al. 2022). Finally, if the program offers not a temporary but a permanent positive income shock, beneficiary households may spend rather than save, which would have positive multiplier effects on local firms and households as in Egger et al. (2022). Rising demand may also raise prices, which would reduce the welfare gains from the program. However, two studies of cash transfers, Cunha, Giorgi, and Jayachandran (2019) in Mexico and Egger et al. (2022) in Kenya, only find sizable price effects in the most remote villages, which are presumably less well integrated to international markets than Addis Ababa.

Our labor market results with partial rollout show that wages rose 14 percent in treated markets and 3 percent in control markets. These results speak to the spatial concentration of the labor market effects of place-based policies in the short run. In the long run, commuting decisions may be more elastic as workers change their commuting patterns to take advantage of higher wages. If this is the case, we would expect the spatial spillovers to be more dispersed if a similar program was only partially rolled out over the long term.

V. Conclusion

Our paper provides a comprehensive evaluation of the UPSNP, Ethiopia's urban public works program, for its intended beneficiaries, the urban poor. We exploit the random rollout of the program across neighborhoods in Addis Ababa, which we combine with detailed survey data on local amenities, employment, and wages. We first compare treated and untreated neighborhoods to present evidence that the program improved local amenities, increased public employment, and crowded out private sector employment.

We then develop a spatial equilibrium model and leverage detailed data on commuting flows to compute the labor market spillovers of the program. We show that, when partially rolled out, it increased wages earned in treated labor markets by 14 percent and wages earned in control labor markets by 3 percent. An estimation strategy that ignores spillovers between neighborhoods would underestimate the equilibrium effects of the program both by underestimating the gains to program neighborhoods and missing entirely the gains to untreated neighborhoods. What is more, it would predict that once rolled out across the city, the program would increase wages by only 9.3 percent, while our model-based estimates suggest that wages increased by 18.6 percent. We then rely on the structure of the model to compute the welfare effects of the program on the urban poor once completely rolled out across the city. We show that welfare gains are six times larger than the direct gains alone once indirect gains from higher private wages and improved amenities are taken into account.

Our results emphasize the importance of taking into account spillover effects in the evaluation of antipoverty programs, and our paper provides a first example of how to do so through a combination of experimentation at scale and spatial modeling. One limitation of our experimental design is that we compare areas that receive the program one year to areas that will receive it the next, which makes it vulnerable to possible anticipation effects in control areas and prevents us from estimating longer-run effects. Our study also suggests that once spillover effects are taken into account, public works deliver higher welfare gains to the urban poor than a cash transfer of similar size. This helps rationalize why many developing countries today implement urban public works and provides evidence in support for extending rural public works programs, such as India's rural employment guarantee, to urban areas (Drèze 2020).

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